

Fractions: Equal Parts Fix-It

Answer Key

Grade 3

Quick Answers

1. $1 \cdot \frac{1}{4}$

2. —

3. (a) = $1 \cdot \frac{1}{4}$, —

4. (a) = 6, (b) = $1 \cdot \frac{1}{6}$, —

5. (a) = 2 cm, —

6. (b) = 10 cm, —

7. (a) = $1 \cdot \frac{1}{2}$, —

8. (b) = 2, (c) = $1 \cdot \frac{1}{12}$, —

What is verified

1 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 2, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 3

3.NF.A – *problems 1, 2, 3, 4, 5, 6, 7, 8*

Difficulty 1–4 of 5 across 16 tagged checks.

1. This rectangle is cut into 4 equal parts, and 1 part is shaded. Write the fraction of the rectangle that is shaded.

Solution: The parts are equal, so a fraction may be named: count the shaded parts over all the parts. There is 1 shaded part out of 4 equal parts, so the shaded fraction is $\frac{1}{4}$

2. Mia cut this circle into three parts and shaded one part. The parts are not all the same size. Say why Mia cannot call the shaded part one third yet.

Model answer (instructor judges): A fraction like one third only names one of three *equal* parts. Mia’s circle is cut into three parts of different sizes, so “1 out of 3 parts” does not tell how much of the circle is shaded. Mia must first cut the circle into three equal parts (or measure with equal units) before naming the shaded amount one third.

3. Rectangle A and Rectangle B are the same size, each cut into four parts with one part shaded. (a) What fraction of Rectangle A is shaded? (b) Circle the rectangle that truly shows one fourth, and tell how you know.

Solution (a): Rectangle A is cut into 4 *equal* parts and 1 is shaded, so the shaded fraction is $\frac{1}{4}$

Model answer (b), instructor judges: Rectangle **A** should be circled. A’s four parts are all the same size, so each part is truly one fourth. B’s parts are different sizes, so B’s shaded part is 1 of 4 parts but *not* one fourth of the whole.

4. Deshawn split the rectangle into three parts using heavy lines and shaded the small part. He says the shaded part is $\frac{1}{3}$ because it is 1 part out of 3. (a) How many small equal squares make the whole? (b) What fraction is really shaded? (c) Name Deshawn’s mistake.

Solution (a): The rectangle is a grid of 2 rows with 3 small squares in each row, so the whole holds $2 \times 3 = 6$ equal squares. 6

Solution (b): The shaded part covers 1 small square out of the 6 equal squares, so the true shaded fraction is $\frac{1}{6}$, not $\frac{1}{3}$. The really-shaded fraction is $\frac{1}{6}$

Model answer (c), instructor judges: Deshawn counted parts that are *not equal*. “1 part out of 3” only makes the fraction one third when the three parts are the same size. Because they are not, he must count the equal small squares instead.

5. Priya drew a number line from 0 to 1 that is 8 cm long, with tick marks that are not evenly spaced; she labeled the first mark $\frac{1}{4}$. (a) How many centimeters apart should the tick marks be to show fourths? (b) Say what was wrong with Priya’s number line.

Solution (a): Fourths cut the whole line into 4 equal lengths. The line is 8 cm long, so each equal length is $8 \div 4 = 2$, meaning the marks belong every 2 cm

Model answer (b), instructor judges: Priya’s tick marks were not evenly spaced, so the pieces of her line were not equal. Fraction marks on a number line must cut the distance from 0 to 1 into *equal* lengths; only then does the first mark show one fourth.

6. Ben folded a strip into two parts and says each part is one half. He measures one part at 12 cm and the other at 8 cm. (a) Is Ben right? Say how you know. (b) If the strip were folded into 2 equal parts, how long would each part be?

Model answer (a), instructor judges: No. Halves must be two *equal* parts, and 12 cm is longer than 8 cm, so the fold did not make halves — neither part is one half of the strip.

Solution (b): The whole strip is $12 + 8 = 20$ centimeters long. Folding into 2 equal parts gives $20 \div 2 = 10$, so each part should be 10 cm

7. Tia’s garden is split into 4 unequal sections; the tomato section covers 4 small squares. Tia says tomatoes take up $\frac{1}{4}$ of the garden because they are 1 section out of 4. (a) What fraction do tomatoes really cover? (b) Explain what Tia got wrong.

Solution (a): The garden grid holds $2 \times 4 = 8$ equal small squares, and the tomato section covers 4 of them: $\frac{4}{8}$ of the garden, which is the same as one half. The true fraction is $\frac{1}{2}$

Model answer (b), instructor judges: Tia counted sections that are not equal in size. Being 1 section out of 4 only means one fourth when all four sections are equal. The equal small squares show tomatoes really cover half the garden — twice as much as Tia claimed.

8. The grid shows 3 rows of 4 small squares. (a) Cut the grid into six equal parts and shade one part. (b) How many small squares are in each equal part? (c) What fraction of the whole rectangle is one small square?

Model answer (a), instructor judges: Any cutting that gives six parts of *equal size* earns credit — for example, six 2-square dominoes (standing or lying down). Each part must cover the same number of small squares, and exactly one part is shaded. Parts may have different shapes as long as their sizes are equal.

Solution (b): The grid holds $3 \times 4 = 12$ small squares shared among 6 equal parts, so

each part gets $12 \div 6 = 2$ squares.

2

Solution (c): One small square is $\frac{1}{12}$ of the 12 equal squares that make the whole, so it is $\frac{1}{12}$

$\frac{1}{12}$